

Q1: Average, minimum, and maximum house prices

```
print("Average price:", df['price'].mean())  
print("Minimum price:", df['price'].min())  
print("Maximum price:", df['price'].max())
```

OUTPUT:

Average price: 4766729.247706422
Minimum price: 1750000
Maximum price: 13300000

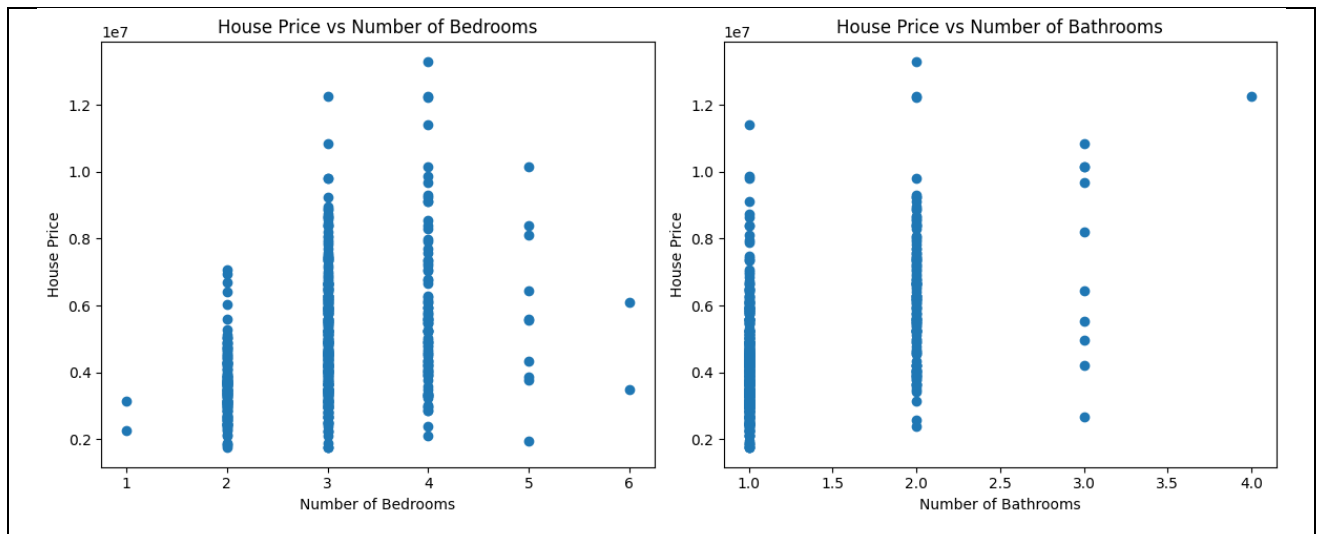
Q2: Correlation with bedrooms, bathrooms; scatter plots

```
print("Price ↔ Bedrooms correlation:", df['price'].corr(df['bedrooms']))  
print("Price ↔ Bathrooms correlation:", df['price'].corr(df['bathrooms']))
```

```
plt.figure(figsize=(12, 5))  
plt.subplot(1, 2, 1)  
plt.scatter(df['bedrooms'], df['price'])  
plt.xlabel('Number of Bedrooms')  
plt.ylabel('House Price')  
plt.title('House Price vs Number of Bedrooms')  
  
plt.subplot(1, 2, 2)  
plt.scatter(df['bathrooms'], df['price'])  
plt.xlabel('Number of Bathrooms')  
plt.ylabel('House Price')  
plt.title('House Price vs Number of Bathrooms')  
plt.tight_layout()  
plt.show()
```

OUTPUT:

Price ↔ Bedrooms correlation: 0.3664940257738689
Price ↔ Bathrooms correlation: 0.517545339455011

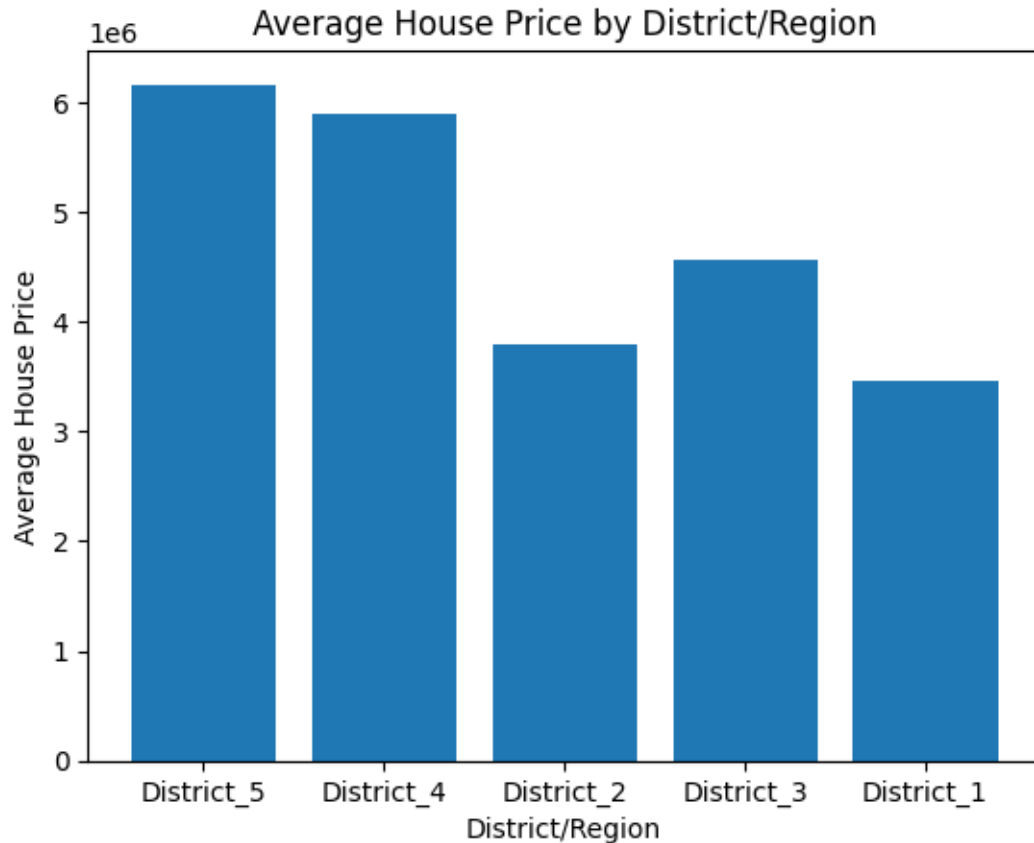


Q3: Average price by district/region

```
print(df.groupby('district')['price'].mean())
plt.figure() # New figure for the bar chart
plt.bar(df['district'].unique(), df.groupby('district')['price'].mean()[df['district'].unique()])
plt.xlabel('District/Region')
plt.ylabel('Average House Price')
plt.title('Average House Price by District/Region')
plt.show()
```

OUTPUT:

```
district
District_1  3.456031e+06
District_2  3.797302e+06
District_3  4.563688e+06
District_4  5.897369e+06
District_5  6.167837e+06
Name: price, dtype: float64
```

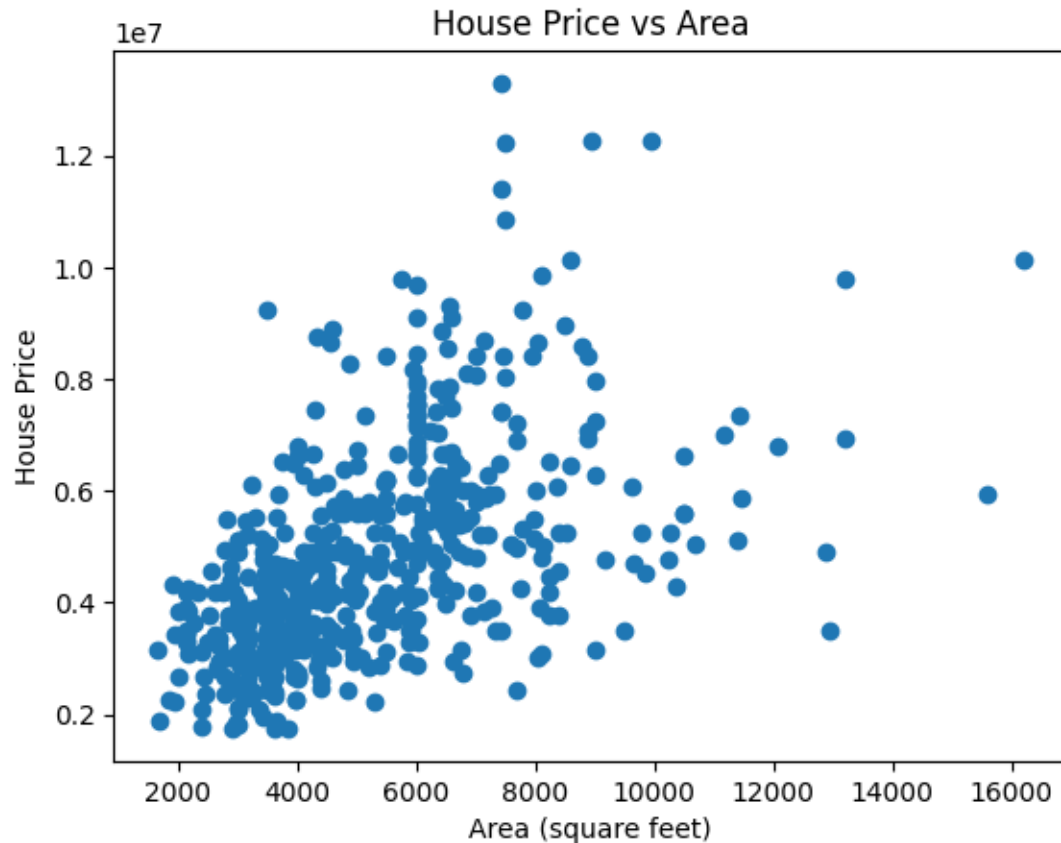


Q4: Relationship between size (area) and price

```
print("Price ↔ Area correlation:", df['price'].corr(df['area']))  
plt.figure() # New figure for the scatter plot  
plt.scatter(df['area'], df['price'])  
plt.xlabel('Area (square feet)')  
plt.ylabel('House Price')  
plt.title('House Price vs Area')  
plt.show()
```

OUTPUT:

Price ↔ Area correlation: 0.5359973457780796



Q5: Outliers in price (boxplot and filtering)

```
plt.figure() # New figure for the boxplot
```

```
sns.boxplot(x=df['price'])
```

```
plt.title('House Price Outliers')
```

```
plt.xlabel('House Price')
```

```
plt.show()
```

```
outliers = df[np.abs(df['price'] - df['price'].mean()) > (3 * df['price'].std())]
```

```
print('Number of outliers:', len(outliers))
```

```
print('Outlier example prices:\n', outliers.head(3)['price'])
```

OUTPUT:

Number of outliers: 6

Outlier example prices:

0 13300000

1 12250000

2 12250000

Name: price, dtype: int64

House Price Outliers

